

SUNIL KUMAR M

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CAREER OBJECTIVE

Instrumentation Engineer with hands-on experience in **industrial instrumentation, preventive and breakdown maintenance, calibration, DCS/PLC-based automation systems, and process control**. Skilled in **field instrumentation, loop checking, troubleshooting**, and ensuring reliable plant operations while adhering to safety standards. Seeking a challenging opportunity in the **Oil & Gas, Petrochemical, or Process Industry** where I can contribute my technical expertise toward safe, efficient, and reliable plant operations while continuously enhancing my professional skills.

PROFESSIONAL EXPERIENCE

Junior Engineer – Instrumentation

Leadec India Pvt. Ltd. (Deputed at Birla Opus Paints – Grasim Paints)

June 2026 – Present

- Perform **preventive maintenance (PM)** of field instrumentation and plant equipment.
- Execute daily **calibration** of **Precia Molen Tank Weighing Systems**.
- Work with **ABB DCS** and **ABB PLC** systems in a continuous process plant.
- Calibrate and troubleshoot **pressure, flow, level, and temperature instruments**.
- Perform **loop checking, signal verification, and breakdown maintenance**.
- Maintain **PM records, calibration reports**, and maintenance documentation.
- Coordinate with production teams while following **plant safety procedures**.

Instrumentation Engineer (Apprentice Trainee)

Piramal Pharma Ltd., Ennore

June 2025 – June 2026

- Installed, calibrated, and maintained **pressure, temperature, flow, and level instruments** in production and utility areas.
- Performed **loop checking** and **functional testing** in accordance with **P&ID** and loop diagrams.

- Executed **compressor field wiring**, including **starting permissives** and **safety interlocks**.
- Implemented **PLC-based compressor start/stop interlock logic** to ensure safe equipment operation.
- Assisted in **PLC control panel wiring**, **I/O termination**, and **field-to-panel signal integration**.
- Supported **SCADA monitoring** and **alarm configuration** for process supervision.
- Performed **calibration of gas analyzers** and ensured accurate analyzer performance.
- Maintained and monitored the **Ambient Air Quality Monitoring System (AQMS)** station.
- Maintained **calibration records**, **maintenance logs**, and compliance documentation as per **GxP** and plant safety requirements.

TECHNICAL SKILLS

- **Field Instrumentation:** Pressure Indicating Transmitters (PIT), Flow Indicating Transmitters (FIT), Level Indicating Transmitters (LIT), Temperature Indicating Transmitters (TIT), RTD (Pt100 – 2-Wire, 3-Wire & 4-Wire), Thermocouples (J & K Type), Pressure Gauges, Control Valves, Turbidity Meter, pH Analyzer, Conductivity Meter, TDS Meter, Gas Analyzers (O₂, Hydrocarbon, VOC, Ammonia, Cyanide), Pressure Switches, Flow Switches, Level Switches, Proximity Sensors.
- **Automation Systems:** PLC (ABB, Siemens, Allen Bradley, Schneider Electric), DCS (ABB), SCADA (Wonderware, WinCC)
- **Industrial Control Concepts:** Interlocks, Permissive Logic, Basic PID Control, Loop Checking, Instrument Calibration
- **Electrical & Panel Work:** Control Panel Wiring, Compressor Interlock Wiring, I/O Termination, Field-to-Panel Signal Integration
- **Documentation:** Calibration Reports, Preventive Maintenance Records, Maintenance Logs, Temperature Mapping, Weighing Scale Qualification

AUTOMATION PROJECTS

Tank Level & Reactor Temperature Control Automation System

- Developed an automated **tank level control system** using **level instrumentation** and **PLC logic** to ensure stable operation.
- Implemented **automatic pump control** with timer-based logic to prevent overflow and reduce manual intervention.

- Designed a **hot water circulation system** to maintain reactor temperature automatically.
- Implemented **PID control logic** to regulate reactor temperature based on **set-point feedback**.
- Integrated an **electro-pneumatic control valve** with a positioner for precise hot water flow control.
- Configured **automatic cut-off** when the set temperature was reached, ensuring process safety and energy efficiency.
- Designed and implemented an **alarm management system** to monitor **agitator run and trip conditions**, improving operational safety and response time.
- Executed **PLC wiring activities** during compressor system upgrades in coordination with vendors, ensuring proper installation, testing, and commissioning.

EDUCATIONAL QUALIFICATION

Bachelor of Engineering (B.E.) – Electronics and Instrumentation Engineering
Hindusthan College of Engineering and Technology, Coimbatore
CGPA: 7.98 | 2021 – 2025

INTERNSHIP

Energim Sustainable Solutions

- Analyzed fertilizer manufacturing processes to improve operational efficiency.

CERTIFICATION

- **Value Added Program on Industrial Automation (PLC, HMI & SCADA)** – Innovative Engineering Products

LANGUAGES

- Tamil
- English